

Erdős Problem #939

EP-939

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Let $r \geq 2$. An r -powerful number n is one such that if $p \mid n$ then $p^r \mid n$.

If $r \geq 4$ then can the sum of $r - 2$ coprime r -powerful numbers ever be itself r -powerful? Are there at most finitely many such solutions?

Are there infinitely many triples of coprime 3-powerful numbers a, b, c such that $a + b = c$?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>